

“Calculus 3”

Multi-Variable Calculus

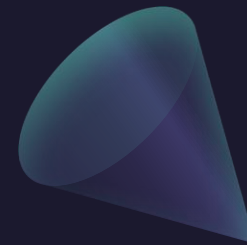
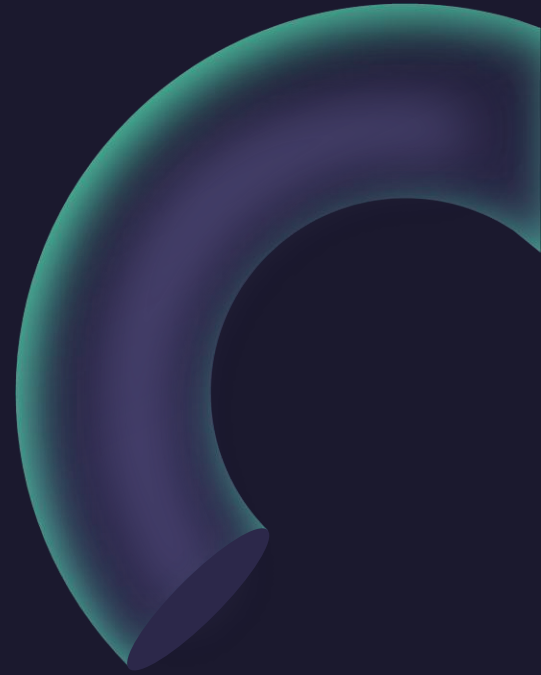
Instructor: Álvaro Lozano-Robledo

Day 27



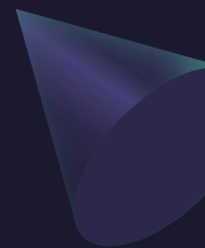
Any Reminders? Any Questions?

- Teaching evaluations are open!
- Exam 3: Friday May 1 – Ch 13.1-13.3 and 16.1-16.8
- Review for Exam 3 during class on Thursday!
- Office hours Thursday 3:30-4:30 at MONT 233
- Calc Night Thursday 6:30-8:30 at MONT 104





ALVARO: Start the recording!



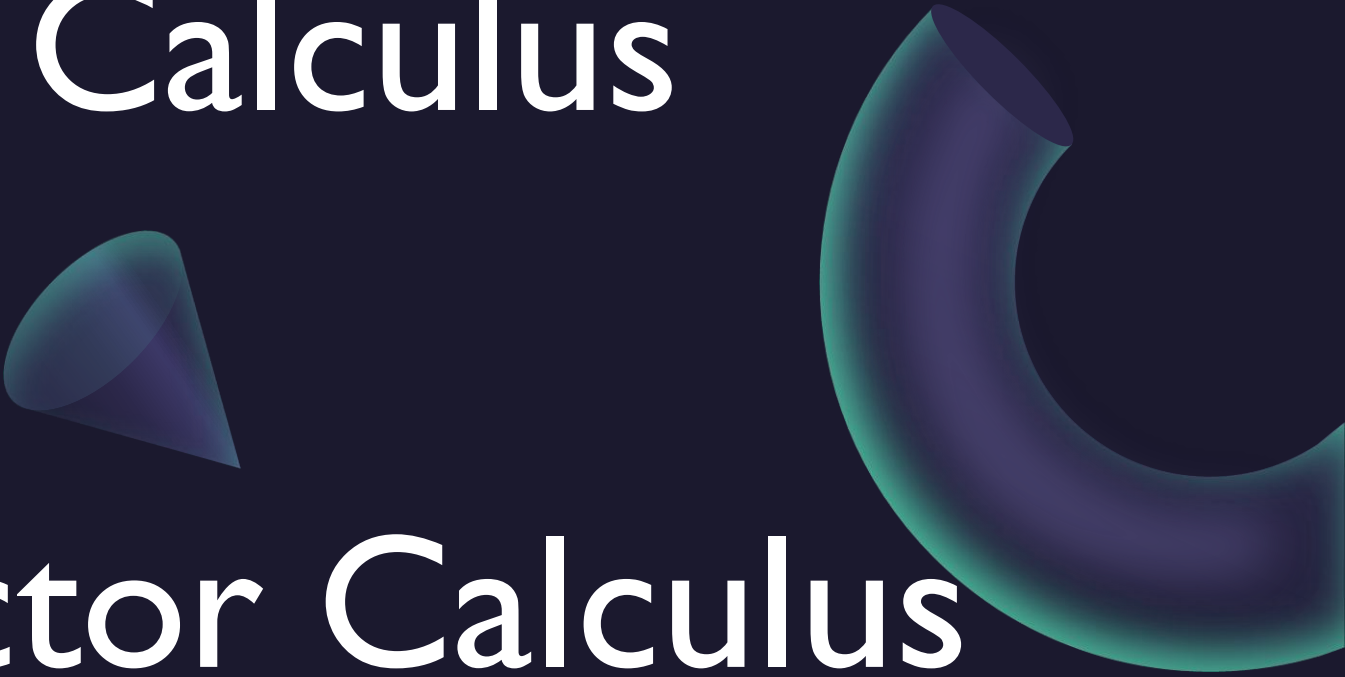
“Calculus 3”

A sphere, a cube, and a cone are positioned in the upper right area of the slide. The sphere is at the top right, the cube is below it, and the cone is further down and to the left.

Multi-Variable Calculus

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Review of Vector Calculus

A cone and a torus are positioned in the lower right area of the slide. The cone is located between the 'Multi-Variable Calculus' and 'Review of Vector Calculus' titles. The torus is on the right side, partially overlapping the 'Review of Vector Calculus' title.

Questions?

1. Vector Functions and Curves

P1. Find the unit tangent vector $\vec{T}(t)$ for $\vec{r}(t) = \langle t, \frac{2}{3}t^{3/2}, t^2 \rangle$ at $t = 1$.

P2. Find the length of the curve $\vec{r}(t) = \langle e^t \cos t, e^t \sin t, e^t \rangle$ for $0 \leq t \leq \ln(2)$.

P3. Consider the curve C formed by the intersection of the cylinder $x^2 + y^2 = 1$ and the plane $z = 1 - y$.

(a) Find a parametrization $\vec{r}(t)$ for C .

(b) Find $\vec{r}'(t)$ and the line integral $\oint_C \langle -y, x, z \rangle \cdot d\vec{r}$.

2. Conservative Fields and Green's Theorem

- P4.** Use Green's Theorem to evaluate $\oint_C (xy + e^{x^2}) dx + (x^2 + \sin(y^2)) dy$ where C is the boundary of the rectangle with vertices $(0, 0)$, $(3, 0)$, $(3, 2)$, and $(0, 2)$.
- P5.** Show that $\vec{F} = \langle y \cos(xy), x \cos(xy) + 2y \rangle$ is conservative and find a potential function f . Use it to evaluate $\int_C \vec{F} \cdot d\vec{r}$ from $(0, 0)$ to $(1, \pi)$.

3. Parametric Surfaces and Surface Area

- P6.** Let S be the part of the plane $3x + 2y + z = 6$ in the first octant.
- (a) Find a parametrization $\vec{r}(x, y)$ for S .
 - (b) Express the domain D as a **Type I (Type- x)** domain.
 - (c) Compute $\vec{r}_x \times \vec{r}_y$ and find the surface area of S using integration.
- P7.** Find the area of the surface $z = xy$ that lies within the cylinder $x^2 + y^2 = 4$.

4. Flux and Vector Integrals

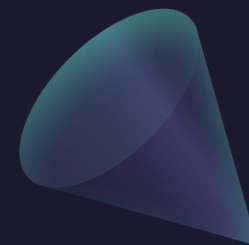
P8. Let S be the part of the cone $z = \sqrt{x^2 + y^2}$ between $z = 0$ and $z = 2$.

- (a) Write a parametrization $\vec{r}(r, \theta)$ for S using **polar coordinates**.
- (b) Find the normal vector $\vec{r}_r \times \vec{r}_\theta$.
- (c) Evaluate the flux of $\vec{F} = \langle x, y, z^4 \rangle$ through S with upward orientation.

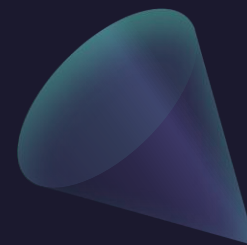
5. Stokes' and Divergence Theorems

P9. Let $\vec{F} = \langle z^2, x^2, y^2 \rangle$. Use Stokes' Theorem to evaluate $\oint_C \vec{F} \cdot d\vec{r}$, where C is the triangle with vertices $(1, 0, 0)$, $(0, 1, 0)$, $(0, 0, 1)$, oriented counterclockwise when viewed from above.

P1. Find the unit tangent vector $\vec{T}(t)$ for $\vec{r}(t) = \langle t, \frac{2}{3}t^{3/2}, t^2 \rangle$ at $t = 1$.

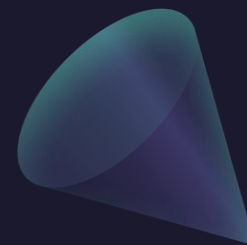


P2. Find the length of the curve $\vec{r}(t) = \langle e^t \cos t, e^t \sin t, e^t \rangle$ for $0 \leq t \leq \ln(2)$.

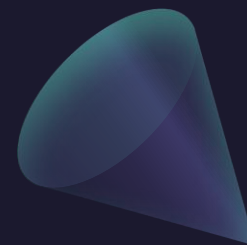


P3. Consider the curve C formed by the intersection of the cylinder $x^2 + y^2 = 1$ and the plane $z = 1 - y$.

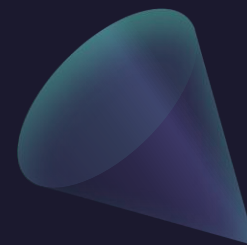
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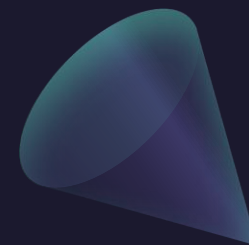


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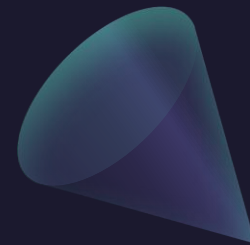


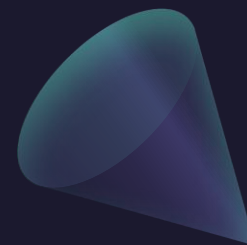
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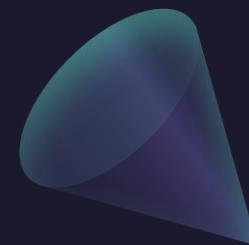
P7. Find the area of the surface $z = xy$ that lies within the cylinder $x^2 + y^2 = 4$.





P8. Let S be the part of the cone $z = \sqrt{x^2 + y^2}$ between $z = 0$ and $z = 2$.

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